



Research and Presenting

06 – Latex Tutorial

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Overview of today

- What is LaTeX?
- Example 1: Basic document
- Where to use LaTeX/Setup
- Basics of LaTeX

What is latex?

- LaTeX (pronounced “LAY-tek” or “LAH-tek”)
- A typesetting language, think a programming language to make documents beautiful
- It is built for anything mathematical related (research papers especially)
- Microsoft word is built using “ WYSWYG - what you see is what you get” mantra of formatted text
- LaTeX is written in plain text with markup tagging
- Why use LaTeX? Have you ever had the feeling when using word or docs that you just couldn’t get something to be in the exact right place? With LaTeX you can.

Example 1: Basic document

```
\documentclass{article} % Starts an article
\usepackage{amsmath} % Imports amsmath
\title{\LaTeX} % Title

\begin{document} % Begins a document
  \maketitle
  \LaTeX{} is a document preparation system for
  the \TeX{} typesetting program. It offers
  programmable desktop publishing features and
  extensive facilities for automating most
  aspects of typesetting and desktop publishing,
  including numbering and cross-referencing,
  tables and figures, page layout,
  bibliographies, and much more. \LaTeX{} was
  originally written in 1984 by Leslie Lamport
  and has become the dominant method for using
  \TeX; few people write in plain \TeX{} anymore.
  The current version is \LaTeXe.

  % This is a comment, not shown in final output.
  % The following shows typesetting power of LaTeX:
  \begin{align}
    E_0 &= mc^2 \\
    E &= \frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}}
  \end{align}
\end{document}
```

$\text{\LaTeX}$

$\text{\LaTeX}$ is a document preparation system for the $\text{\TeX}$ typesetting program. It offers programmable desktop publishing features and extensive facilities for automating most aspects of typesetting and desktop publishing, including numbering and cross-referencing, tables and figures, page layout, bibliographies, and much more. $\text{\LaTeX}$ was originally written in 1984 by Leslie Lamport and has become the dominant method for using $\text{\TeX}$; few people write in plain $\text{\TeX}$ anymore. The current version is $\text{\LaTeX} 2_{\epsilon}$.

$$E_0 = mc^2 \quad (1)$$

$$E = \frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (2)$$

How to setup Latex?



Online

LaTeX online services like [Overleaf](#), [Papeeria](#), [CoCalc](#) or [Prism](#)



Linux

[TeX Live](#) distribution



Mac OS

[MacTeX](#) distribution



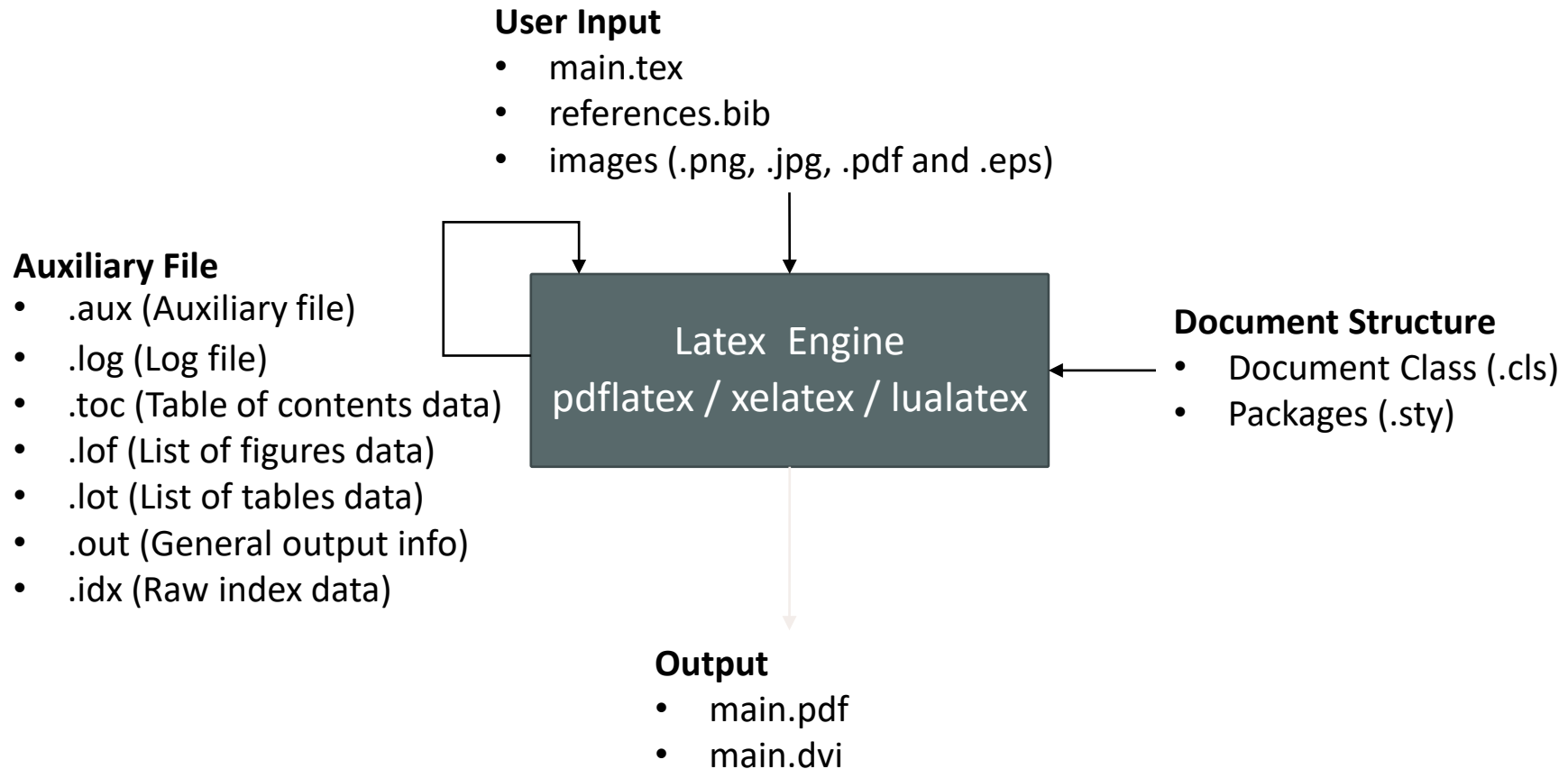
Windows

[MiKTeX](#) or [TeX Live](#) distributions

LaTeX editor

[TeXStudio](#)

How Latex Work



Basics of latex: important chars

- There are some important characters and a characteristic of LaTeX
- Characters:
 - “\”: Starts a typesetting command
 - “{}”: Any arguments are put inside the curly braces
 - “\\”: Newline without a new paragraph
 - “%”: Used to start a comment, everything after on the same line is treated as such
- Characteristic:
 - LaTeX wraps adjacent lines as if they were part of a same paragraph.
Insert extra return for a new paragraph

Source:

This is one paragraph.

This is another.

Output:

This is one paragraph.

This is another.

Basic Document Structure

- The format of a document is pretty simple.
- In the preamble
 - document class
 - packages
- In the front matter
 - Title/author
 - Table of contents/ List of figures
- In the body
 - Contents
- In the back matter
 - bibliography

Basics of latex: document class

- Document classes is how LaTeX interprets a particular document
- Main available types are:
 - **Article:** articles in scientific journals, short reports, program documentation
 - **Report:** longer reports containing several chapters, small books, thesis work
 - **Slides/beamer:** for presentations
- Starting a new document:

```
\documentclass [options] {style}
```

```
\begin{document}
```

```
%your writing here
```

```
\end{document}
```

Predefined classes like
[article](#), [report](#), [letter](#) or [book](#)

Options within a class such as
[Paper size](#), [font size](#) etc

Basics of latex: Packages

- Just like any other programming language, people can build packages that give the programming language to do certain things and other people can download said packages.
- Most packages are already installed by default
- To use them, do a `\usepackage[options]{packagename}` before you begin your document

Basics of latex: Sectioning

- Sectioning is to create different sections and subsections in your documents.
 - `\section{name of section}`: creates a section with the specific name, is numbered
 - `\subsection{name of subsection}`: creates a subsection under the larger section
 - `\subsubsection{name of subsubsection}`: I think you get the idea
 - `\section*{Name of section}` creates a section with the specific name, without the number.
 - `\paragraph{name of paragraph}`: like section but not numbered
 - `\subparagraph{name of subparagraph}`: like paragraph but indented, goes under a paragraph

Basics of latex: Sectioning

- `\section{name of section}`
- `\subsection{name of subsection}`
- `\subsubsection{name of subsubsection}`
- `\paragraph{name of paragraph}`
- `\subparagraph{name of subparagraph}`
- `\section*{Name of section}`

2 Name of section

creates a section with the specific name, is numbered

2.1 Name of subsection

creates a subsection under the larger section

2.1.1 Name of subsubsection

I think you get the idea

Name of paragraph like section but not numbered

Name of subparagraph like paragraph but indented, goes under a paragraph

Name of section

creates a section with the specific name, without the number.

Basics of Latex: Tables

- Simple tables can be produced by `\begin{tabular}[pos]{tablespec}`
- Within the `{tablespec}` section, one details the number of columns, the alignment, and the number of vertical lines of the table. `{lrc}`, `{||r|c}`
- Then type in from left to right, the values for each cell with `&` in between.
- Put `"\\`" at the end of each row, then input another row of values if needed.
- `\hline`

```
\section{Table}
Here is table~\ref{tab:table1} in the text.
```

```
\begin{table}[h!]
\begin{center}
\caption{Your first table.}
\label{tab:table1}
\begin{tabular}{l|c|r} % <-- Alignments: 1st column left, 2nd middle and 3rd right, with vertical lines in between
\textbf{Value 1} & \textbf{Value 2} & \textbf{Value 3}\\
 $\alpha$  &  $\beta$  &  $\gamma$  \\
\hline
1 & 1110.1 & a\\
2 & 10.1 & b\\
3 & 23.113231 & c\\
\end{tabular}
\end{center}
\end{table}
```

Here is table 1 in the text.

Table 1: Your first table.		
Value 1	Value 2	Value 3
α	β	γ
1	1110.1	a
2	10.1	b
3	23.113231	c

Basics of Latex: Image

- Using the *figure environment* and the *graphicx package*
- `\usepackage{graphics}`

h (here) – same location
t (top) – top of page
b (bottom) – bottom of page
p (page) – on an extra page
! (override) – will force the specified location

```
\begin{figure}[t]
  \centering
  \includegraphics[width=0.5\linewidth]{cat.jpg}
  \caption{A Cute Cat.}
  \label{fig:boat1}
\end{figure}
```

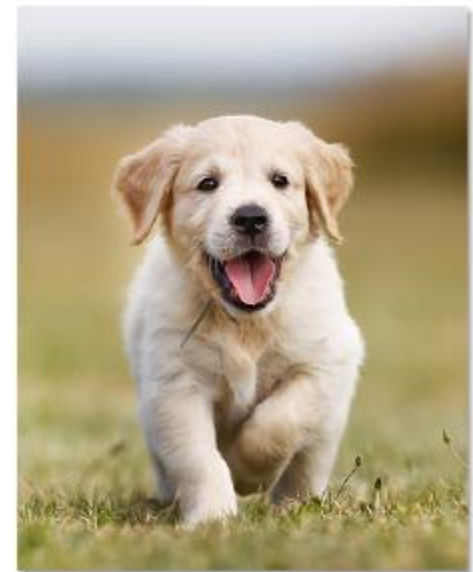


Figure 1: A Cute Cat.

Basics of Latex: Equation

- allows two writing modes for mathematical expressions: the **inline** math mode and **display** math mode

```
\section{Equation}
```

An inline equation: $a^2 + b^2 = c^2$.

```
\begin{equation}
  g_{\gamma_n}(t) =
  \frac{1}{\sqrt{s_n}} g\left(\frac{t-p_n}{s_n}\right) \exp\left\{j(2\pi f_n t + \phi_n)\right\},
  \label{eq1}
\end{equation}
```

```
\begin{align}
  E_0 &= mc^2 \\
  E &= \frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}}
\end{align}
```

An inline equation: $a^2 + b^2 = c^2$.

$$g_{\gamma_n}(t) = \frac{1}{\sqrt{s_n}} g\left(\frac{t-p_n}{s_n}\right) \exp\left\{j(2\pi f_n t + \phi_n)\right\}, \quad (1)$$

$$E_0 = mc^2 \quad (2)$$

$$E = \frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (3)$$

Basics of Latex: lists

- Can create ordered or unordered lists or sublists as shown on the right.

```
Unordered list:  
\begin{itemize}  
  \item First item  
  \item Second item  
\end{itemize}
```

```
Ordered list:  
\begin{enumerate}  
  \item First item  
  \item Second item  
  \item Third item  
\end{enumerate}
```

Unordered list:

- First item
- Second item

Ordered list:

1. First item
2. Second item
3. Third item

Basics of Latex: Citations

- `\cite{bibtexkey}`, `citeyear{bibtexkey}`
- It is more convenient to create a bibliography file, called BibTeXfile(.bib), and use it as needed.
- JabRef (<http://jabref.sourceforge.net/>)

The screenshot shows the JabRef application window titled "mybibfile-others.bib - JabRef". The interface includes a menu bar (File, Edit, Library, Quality, Lookup, Tools, View, Help), a search bar, and a sidebar with "Groups" and "Filter groups...". The main area displays a table of bibliography entries.

Entrytype	Author/Editor	Title	Year	Journal/Booktitle	Star
InProceedings	Sawal and Wong	Stuck-at Faults in ReRAM Neuromorphic Circuit Array and their Correction throug...	2024	Latin American EL...	★ ★ ★ ★
Article	Shin et al.	Fault-Free: A Framework for Analysis and Mitigation of Stuck-at-Fault on Realistic ...	2023	IEEE Transactions ...	★ ★ ★ ★
InProceedings	Schroedter et al.	Fault Impact Map for Memristive Crossbar Neural Networks	2024	International Con...	★ ★ ★ ★
Article	Weng and others	FKeras: A Sensitivity Analysis Tool for Edge Neural Networks	2024	ACM Journal on ...	★ ★ ★ ★
InProceedings	Chen et al.	BinFi: an efficient fault injector for safety-critical machine learning systems	2019	International Con...	★ ★ ★ ★
Article	Li and others	Analogue Signal and Image Processing with Large Memristor Crossbars	2017	Nature Electronics	★ ★ ★ ★

Below the table, there are tabs for "Required fields", "Optional fields", "Other fields", "Comments", "General", "Abstract", "Citation information", "Citation relations", "Related articles", "BibTeX source", and "LaTeX Citations". The "Required fields" tab is active, showing fields for Author, Title, Journal, Year, and Citationkey.

Article (Shin2023)
Shin, Hyein / Kang, Myeonggu / Kim, Lee Sup
Fault-Free: A Framework for Analysis and Mitigation of Stuck-at-Fault on Realistic ReRAM-based DNN Accelerators
2023
IEEE Transactions on Computers, Vol. 72
p. 2011-2024

Basics of Latex: Citations

The screenshot shows a LaTeX editor with a project structure on the left. The main window displays the content of `mybibfile-others.bib`, which contains three entries:

```
@article{Ibrahim2020,
  title = {Soft Errors in DNN Accelerators: A Comprehensive Review},
  author = {Younis Ibrahim and others},
  journal = {Microelectronics Reliability},
  pages = {113969},
  volume = {115},
  year = {2020}
}

@article{Guo2019,
  title = {Unsupervised Learning on Resistive Memory Array Based Spiking Neural Networks},
  author = {Yilong Guo and Huaqiang Wu and Bin Gao and He Qian},
  journal = {Frontiers in Neuroscience},
  pages = {812},
  volume = {13},
  year = {2019}
}

@inproceedings{Pedretti2021,
  title = {Conductance Variations and their Impact on the Precision of in-Memory Computing with Resistive Switching Memory (RRAM)},
  author = {Giacomo Pedretti and Elia Ambrosi and Daniele Ielmini},
  booktitle = {IEEE International Reliability Physics Symposium (IRPS)},
  city = {Monterey, CA, USA},
  month = {3},
  pages = {1-8},
  year = {2021}
}
```

On the right, the `References` section of the compiled document is shown, displaying the formatted citations for the three entries.

The screenshot shows a LaTeX editor with a project structure on the left. The main window displays the content of `GbSA-Blind-version.tex`, which includes a section titled `\section{Introduction}` and a paragraph of text:

```
\maketitle

\section{Introduction}\label{Intro}
Deep neural networks (DNNs), and in particular convolutional neural networks (CNNs), have achieved high attention and great performance in classification, detection, and recognition tasks in various domains such as healthcare, autonomous driving, space applications, and other safety-critical systems~\cite{Ibrahim2020}. Their success is primarily due to their ability to extract complex features through deep, multi-layered architectures, which is based on computationally intensive vector-matrix multiplication (VMM). To achieve high accuracy, the complexity of CNNs increases, which in turn leads to significant growth in computational cost, memory requirements, and energy consumption.

ReRAM devices, fabricated at the nanoscale, can be seamlessly integrated into crossbar arrays, providing an efficient solution for implementing large numbers of synaptic connections in neural networks~\cite{Guo2019, Pedretti2021} and graph processing~\cite{Ghasemi2022}. ReRAM crossbars can perform massively memory-intensive VMM operations in parallel and have been widely adopted in accelerators for CNN such as \textit{ISAAC}~\cite{Shafiee2016}, \textit{PRIME}~\cite{Chi2016}, and \textit{PipeLayer}~\cite{Song2017}.
```

On the right, the `References` section of the compiled document is shown, displaying the formatted citations for the three entries.

Advanced Latex: circuit diagrams/syntax highlighting

- Include this because most of us here are EE/Compe majors
- Can use the **circuitikz** package to create circuit diagrams
- Can use the listings package to use syntax highlighting for various languages that you include code snippets in your document.

```
1. \begin{circuitikz}
2.   \draw (0,0)
3.     to[V,v=$U_q$] (0,2) % The voltage source
4.     to[short] (2,2)
5.     to[R=$R_1$] (2,0) % The resistor
6.     to[short] (0,0);
7.   \draw (2,2)
8.     to[short] (4,2)
9.     to[L=$L_1$] (4,0)
10.    to[short] (2,0);
11.   \draw (4,2)
12.     to[short] (6,2)
13.     to[C=$C_1$] (6,0)
14.     to[short] (4,0);
15. \end{circuitikz}
```

This would give us the following diagram:

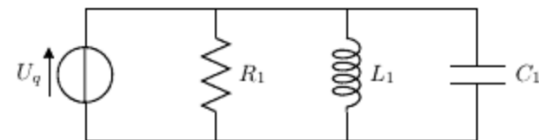


Figure 1: My first circuit.

Persian Packages in LaTeX

- **xepersian** (Recommended and Standard Package)
 - Full Persian support: Automatically handles right-to-left text, Persian numerals, Persian footnotes, and the Persian calendar
 - System fonts: Allows the use of system fonts such as B Nazanin, etc
 - Requires **XeLaTeX**: Documents must be compiled with the XeLaTeX engine
- **Babel** (Modern and International Approach)
 - International standard: Works well with modern engines such as LuaLaTeX
 - Good for multilingual projects: Suitable when documents contain multiple languages.
 - Less optimized for Persian users: Persian configuration is not as ready-to-use as xepersian.
- **ArabTeX (Old and Largely Obsolete)**
 - An older system for typesetting Arabic-script languages, including Persian.
 - Difficult to use and outdated

An Example of Persian XePersian

```
\documentclass{article}
\usepackage{graphicx}

\usepackage{xepersian}
\settextfont{B Nazanin}
\setlatintextfont{Times New Roman}
% \settextfont{B Zar}

\title{عنوان فارسی}

\begin{document}
  \maketitle
  \begin{abstract}
    این متن در بخش چکیده است.
  \end{abstract}

  \section{مقدمه}
  هدف از این درس آشنایی و کسب مهارت در اصول و روش‌های انجام تحقیق، اصول تهیه انواع ارائه‌های نوشتاری، مسائل مربوط به انواع ارائه‌های شفاهی و آشنایی با ابزارهای مربوطه می‌باشد.

  \section{روشها}
  \subsection{روش پیشنهادی اول}
    متن روش پیشنهادی اول.

  \subsection{روش پیشنهادی دوم}
    متن مربوط به روش پیشنهادی دوم با نام انگلیسی \text{ABC}.

  \begin{latin}
    \section{Abstract}
    This is a paragraph written in English to show how xepersian handles multi-language documents. Notice how the alignment shifts to the left.
  \end{latin}

  دوباره به فارسی برگشتیم. شماره‌گذاری‌ها هم خودکار فارسی می‌شوند: ۱، ۲، ۳.
\end{document}
```

عنوان فارسی

۲۸ اردیبهشت ۱۴۰۵

چکیده

این متن در بخش چکیده است.

۱ مقدمه

هدف از این درس آشنایی و کسب مهارت در اصول و روش‌های انجام تحقیق، اصول تهیه انواع ارائه‌های نوشتاری، مسائل مربوط به انواع ارائه‌های شفاهی و آشنایی با ابزارهای مربوطه می‌باشد.

۲ روشها

۱.۲ روش پیشنهادی اول

متن روش پیشنهادی اول.

۲.۲ روش پیشنهادی دوم

متن مربوط به روش پیشنهادی دوم با نام انگلیسی ABC.

3 Abstract

This is a paragraph written in English to show how xepersian handles multi-language documents. Notice how the alignment shifts to the left.

دوباره به فارسی برگشتیم. شماره‌گذاری‌ها هم خودکار فارسی می‌شوند: ۱، ۲، ۳.

Thank You